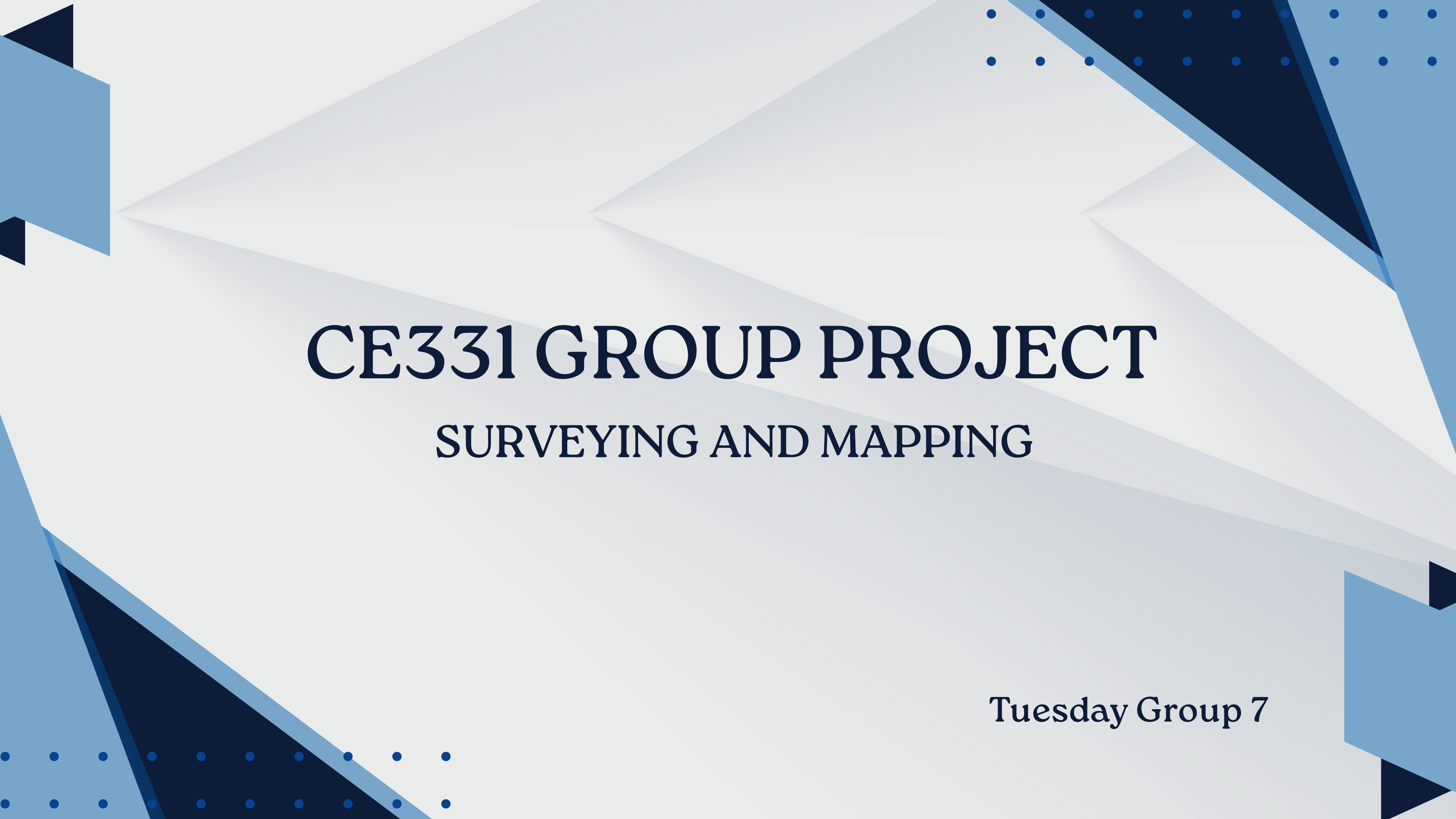




CE331 GROUP PROJECT

SURVEYING AND MAPPING

Tuesday Group 7

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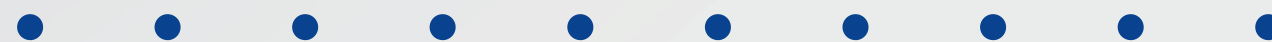
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INTRODUCTION

This project and its presentation is an integral part of the Principles of Geoinformatics course (CE331)

- **Significance:** A hands-on end-to-end experience where we apply concepts taught in class and procedures taught in the laboratory sessions.
- This project involves all necessary steps for **map-making**, including setting up a **control network**, **levelling**, **traversing**, **feature observations**, and finally collating all the collected data into a map of the area allotted to us.



OBJECTIVES

- **Apply principles taught in class:** Follow the steps of map-making taught in class from scratch.
- Gain hands-on experience with geoinformatics **tools and software**, including Auto-level, Total Station, and QGIS.
- **Data Processing:** Learn to analyze and interpret geospatial data, including traversal, and report the Quality of Work of the data collected.
- Create detailed and precise maps.



METHODOLOGY

Reconnaissance

Levelling

Traversing

Feature Mapping

Creation of Map



Week 1- Reconnaissance

We began by exploring the assigned area, After some careful searching we have identified five change points and marked them with paint

Week 3 & 4 - Travesing

We spent two weeks doing travesing to measure the angles and distances between the control points

WORK-FLOW

Week 2- Leveling

We used an auto level to carry out leveling on the five Change Points , taking the benchmark near the GI lab for our reference

Week 5 & 6- Feature Mapping

In the final two weeks, we worked on feature mapping at all the five CP carefully marking the locations of trees, lamps, electric poles and the buliding to create a clear and infromative map.

Reconnaissance

Before establishing our control network, we **explored** the survey area.

This helped us in setting up **control points** at the most suitable locations, considering

- Features in the area
- Inter-visibility
- Points being open to the sky

We finalized **5** control points, all necessary for plotting the various features present in the area, including trees and infrastructure.



Levelling

Before starting the detailed survey, we conducted levelling to determine the relative heights of different points in the area.

This process helped us establish a benchmark and maintain accuracy in plotting the terrain profile.

Key Steps:

- Selected a **benchmark (BM)** near the GI lab.
- Performed **levelling** to transfer the RLs (Reduced Levels) to all control points.
- Ensured proper **back sight** and **fore sight** readings for error minimization.
- Applied **arithmetical checks** to verify the accuracy of observations.

Outcome:

A precise elevation network was established, ensuring reliable height data for contouring and mapping.



Data Sheet

BS	FS	RISE	FALL	RL	REMARK	US(BS)	LS(BS)	US(FS)	LS(FS)	Distance (m)			Cumulative Distance	Correction(m)	Adjusted RL(m)
										D1	D2	Total	(m)		
BM	1.078			128.409	BM	1.138	1.018			12			0	0	128.409
A	1.468	1.4	0.322	128.087		1.558	1.37	1.433	1.312	18.8	12.1	24.1	24.1	0.000666666666	128.0876667
CP-4	1.3	1.302	0.166	128.253	CP4	1.369	1.231	1.413	1.195	13.8	21.8	40.6	64.7	0.001333333333	128.2543333
CP-3	1.431	1.342	0.042	128.211	CP3	1.519	1.348	1.414	1.273	17.1	14.1	27.9	92.6	0.002	128.213
CP-2	1.333	1.482	0.051	128.16	CP2	1.465	1.21	1.604	1.365	25.5	23.9	41	133.6	0.002666666667	128.1626667
B	1.312	1.445	0.112	128.048		1.419	1.205	1.565	1.325	21.4	24	49.5	183.1	0.003333333333	128.0513333
CP-1	1.325	1.315	0.003	128.045	CP1	1.491	1.159	1.424	1.205	33.2	21.9	43.3	226.4	0.004	128.049
CP-5	1.384	1.404	0.079	127.966	CP5	1.465	1.3	1.55	1.259	16.5	29.1	62.3	288.7	0.004666666667	127.9706667
D	1.37	1.287	0.097	128.063		1.405	1.355	1.372	1.203	5	16.9	33.4	322.1	0.005333333334	128.0683333
BM		1.03	0.34	128.403	BM			1.089	0.97		11.9	16.9	339	0.006	128.409
SUM	12.001	12.007	0.603	0.609									339		
LHS	= $\sum BS - \sum FS =$		-0.006	error(m)	-0.006					Quality of work	c				
RHS	= $\sum Rise - \sum Fall =$		-0.006	correction(m)	0.006	Misclosure= 6mm				ordinary	100				
RHS	= Last RL - First RL =		-0.006	k(km)	0.339					Rough	24				
				$E=c(k)^{1/2}$	6					Accurate	12				
				c	10.3050808					Precise level	4				
				Quality: Accurate!						High Quality	1				

TRAVERSING

Traversing was carried out to measure the **angles and distances** between control points for accurate mapping of the area.

Procedure:

- The **Total Station** was **centred and levelled** precisely at each change point.
- **Bipod** and **prism** were used to take observations between consecutive points.
- **Horizontal** and **vertical angles** along with **slope** distances were recorded.
- Data was processed to compute **coordinates** and ensure **closing error** checks.

Outcome:

An accurate traverse framework was established, forming the base for detailed feature plotting and contour generation.



Data Sheet

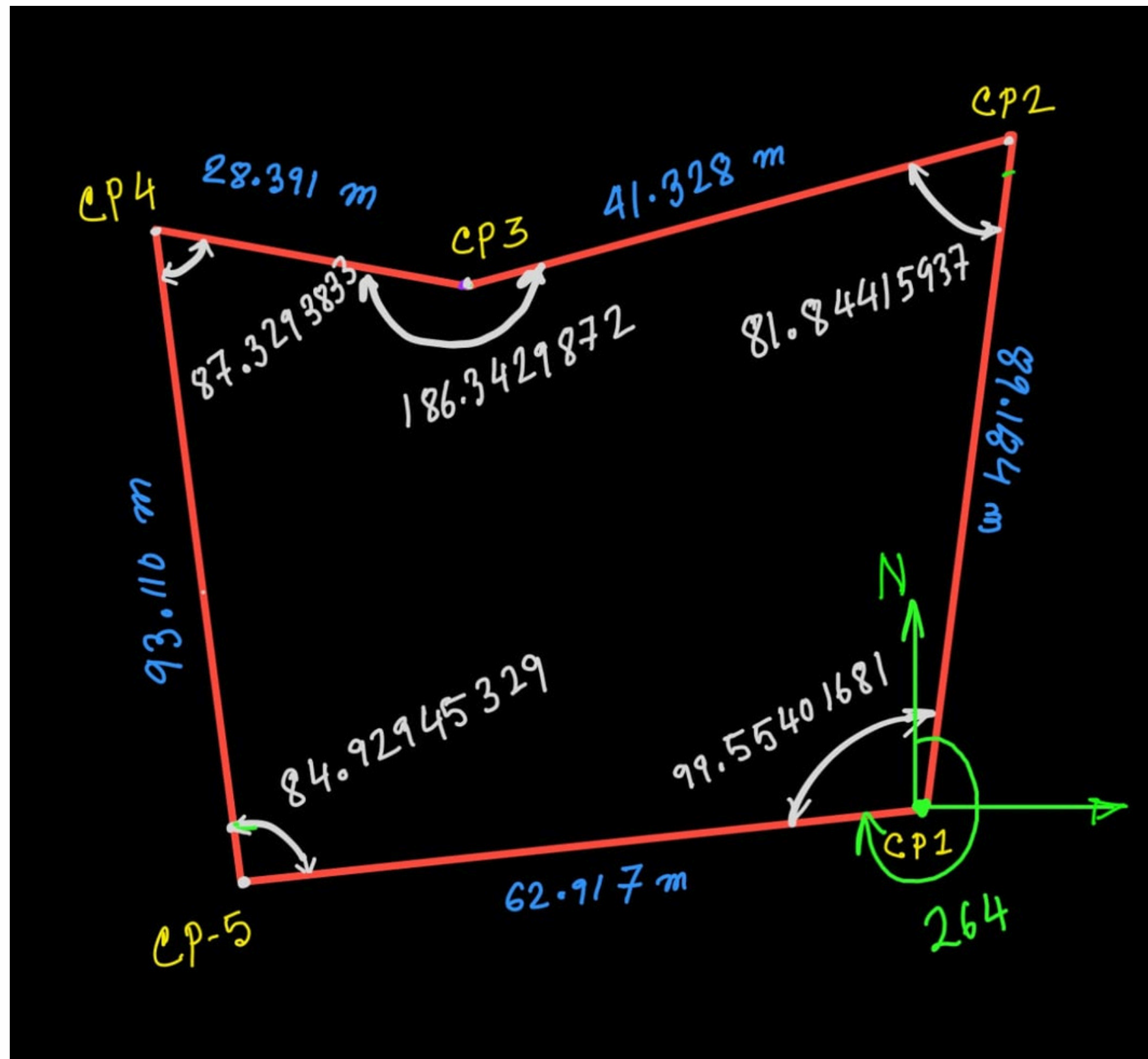
CP	Station Observed	Face (L/R)	Horizontal Angle					
			Readings	Distance	Angle In Decimal Format	Internal Angle(°)	Correction (°)	Corrected Internal Angle
2	1	R	0° 0' 0"	89.184	0	81.83910352	0.00505585241	81.84415937
	3		81° 50' 21"	41.328	81.83916667			
	1'		0° 0' 1"	89.184	0.000277777778			
	Error		0° 0' 1"		0.000277777778			
3	2	R	0° 0' 00"	41.328	0	186.3379313	0.00505585241	186.3429872
	4		186° 20' 15"	28.391	186.3375			
	2'		359° 59' 57"	41.328	359.9991667			
	Error		0° 0' 3"		0.000833333333			
4	3	R	0° 0' 0"	28.391	0	87.32432749	0.00505585241	87.32938334
	5		87° 19' 22"	93.11	87.32277778			
	3'		359° 59' 37"	28.391	359.9936111			
	Error		0° 0' 23"		0.006388888889			
5	4	R	0° 0' 0"	93.11	0	84.92439744	0.00505585241	84.92945329
	1		84° 55' 25"	62.917	84.92361111			
	4'		359° 59' 48"	93.11	359.9966667			
	Error		0° 0' 12"		0.003333333333			
1	5	R	0° 0' 0"	62.917	0	99.54896095	0.00505585241	99.55401681
	2		99° 33' 17"	89.184	99.55472222			
	5'		0° 01' 15"	62.917	0.02083333333			
	Error		0° 01' 15"		0.02083333333			
					Sum of all internal angles ---> (Uncorrected)	539.9747207	0.02527926205	540
	Quality of work	c	Permissible limit of closing error		Actual Sum of Angles ----> {(n-2)*180 }	540		
	First Order	6	13.41640786		Angle Misclosure (in degree) -->	-0.02527926205		
	Second Order	15	33.54101966		Angle Misclosure (in seconds) -->	91.00534336		
	Third Order	30	67.08203932					

Data Sheet

	WCB	Internal Angle		Latitude			Departure		
Line	(in degrees)	(in degree)	Length(m)	Calculated	Correction	Corrected	Calculated	Correction	Corrected
1-2	183.5540168	99.55401681	89.184	5.517541	0.010933	5.528474	89.02134547	-0.00899437	89.0123511
2-3	265.3981762	81.84415937	41.328	-41.199838	0.005066	-41.194772	-3.31166419	-0.004168004847	-3.315832195
3-4	271.7411634	186.3429872	28.391	-28.3813716	0.00348	-28.3778916	0.8659225692	-0.002863284592	0.8630592846
4-5	179.0705467	87.32938334	93.11	1.4990484	0.011415	1.5104634	-93.09449358	-0.00939031	-93.10388389
5-1	264	84.92945329	62.917	62.5260131	0.0077131	62.5337262	6.55065099	-0.0063452952	6.544305695
	sum--->		314.93	0.0386071	0.0386071	0	-0.03176127	-0.03176127	0
					0				

Data Sheet

CP	Easting	Northing
CP1	1000	1000
CP2	1005.517541	1089.003357
CP3	964.328636	1085.692481
CP4	935.95233	1086.556416
CP5	937.454859	993.446077



Feature Mapping

Feature mapping was conducted to record and locate all important details within the survey area.

Procedure:

- The **Total Station** was **centred and levelled** accurately at each station.
- Using a **bipod** and **prism**, we measured the coordinates of key features.
- Mapped features included **trees, roads, manholes, electric poles, and buildings.**
- All observations were stored electronically for **processing and plotting** in mapping software.

Outcome:

A detailed topographic map was generated, showing both natural and man-made features with high positional accuracy.



Creation of Map

Finally map is created on QGIS Software

Procedure

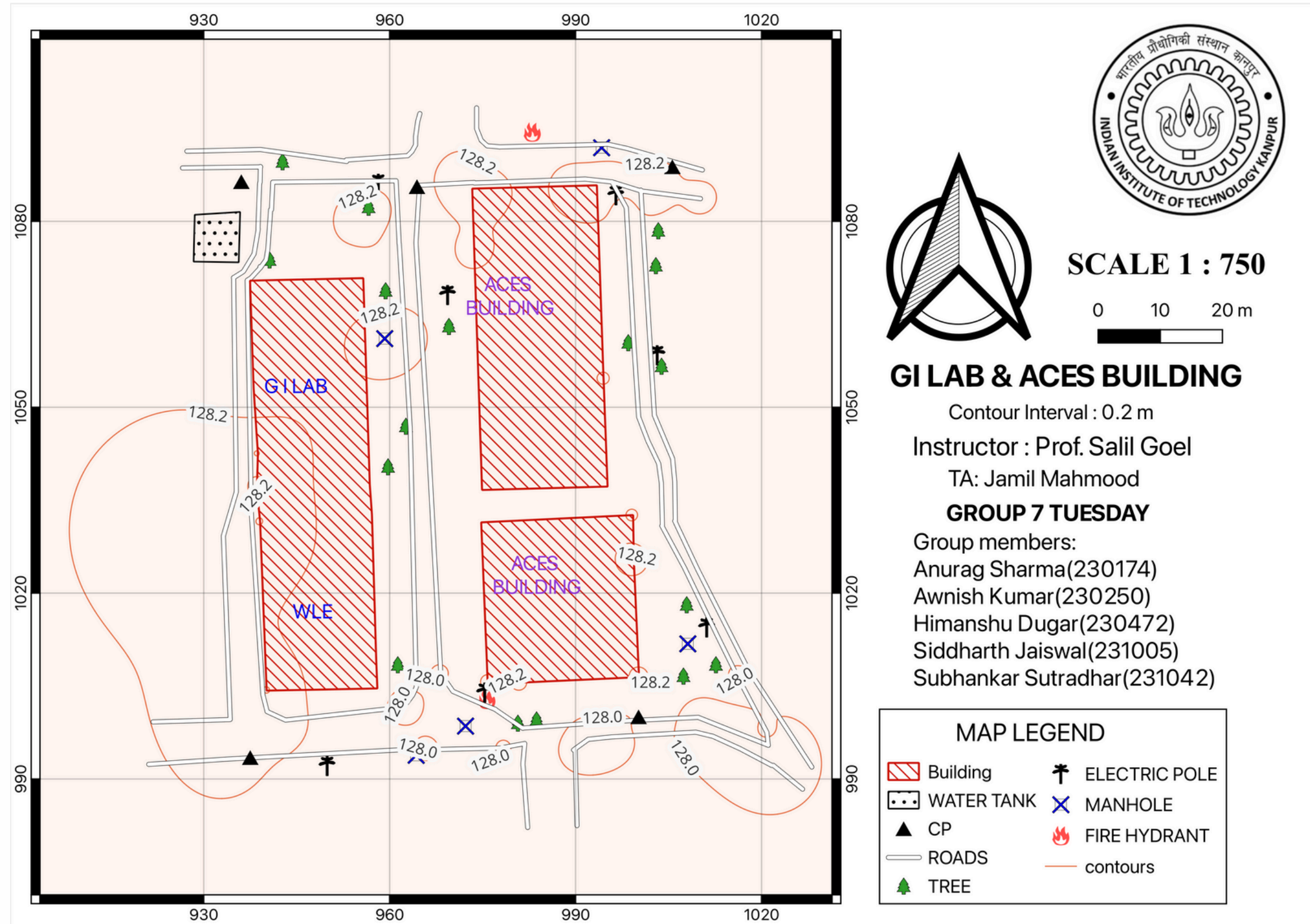
- The survey observations collected using the Total Station were first downloaded and organized.
- The collected point data was imported into QGIS software.
- Separate vector layers (point, line, and polygon) were created to represent different features on the map.
- Each feature was assigned proper symbology and labels (e.g., buildings, water tanks, roads, trees, electric poles, manholes).
- Contour lines were added to represent elevation differences across the site.
- The Layout Manager was used to design the final map.

Outcome:

A clear and accurate topographic map of the GI Lab and ACES Building area was successfully generated using QGIS.



Final Map



Conclusion

Key Takeaways

- **Result:** A detailed map of the Civil Department Building, ACES Building, and their surrounding areas.
- Gained **practical experience** in:
 - Setting up and operating surveying instruments.
 - Recording and processing field observations.
 - Handling real-world constraints in data collection.
 - Understanding the complete survey-to-map workflow.

Challenges Encountered

- In lab 2 **Control points** (CPs) were **erased** due to rain and vehicle movement.
- Difficulty in **centring and levelling** at one CP because of **loose soil** conditions.
- **Vehicles parked** at control points interrupted **traversing and levelling**.
- **Limited time** was available to complete the project.





THANK
YOU

Referencing

All the references used in preparing this Project report are as follows:

- maps.google.com
- Lab reports
- QGIS software for map making.
- Lecture notes by Professor Salil Goel.